Optical Character Recognition (OCR) of Low-Resource Languages

Lightweight OCR for North Sámi Heritage Digitization

Jan Paweł Musioł

Supervised by Aparna S. Varde & Juan Heredia

The Maersk Mc-Kinney Møller Institute
University of Southern Denmark

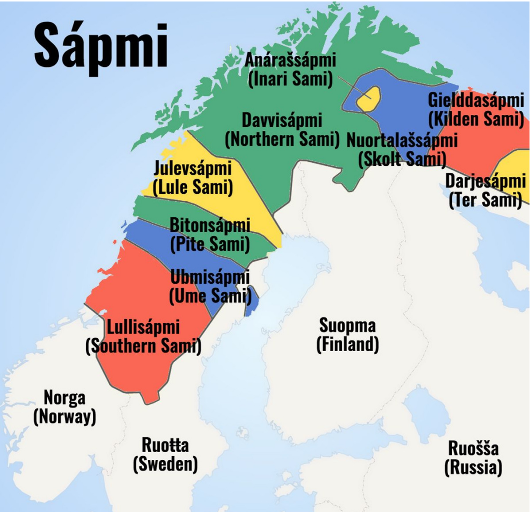
Bachelor Thesis 2026 • IISA 2026 full paper • SCAI 2026 abstract

June 2026

Outline

What are Sámi languages & Why this matters

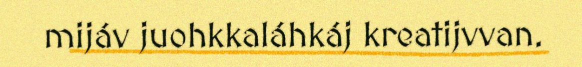
- Nine mutually unintelligible Sámi languages across Northern Europe
- North Sámi has ∼25,000 speakers
- classified by UNESCO as definitely endangered
- Heritage preservation is critical for culture survival → digitization/OCR makes it possible
- Technology should be available for everyone



What is OCR?

OCR: Optical Character Recognition

- Task of converting scanned documents into machine readable text
- Essential for digitization and preservation of culture
- Archives, libraries


mijáv juohkkaláhkáj kreatijvvan.

↓

`mijáv juohkkaláhkáj kreatijvvan.`

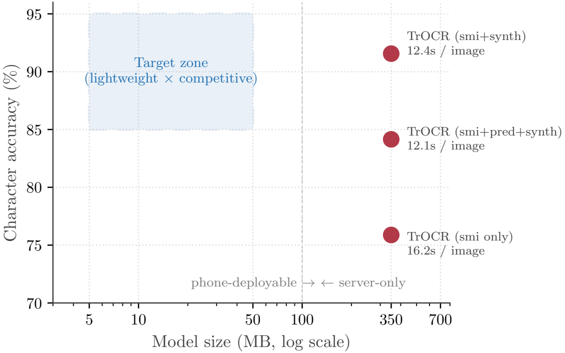
The gap

Comparative study: Tessearct vs Transcribus vs TrOCR.[1]

State of the art:
Transformer-based OCR (TrOCR)

- 91.45% character accuracy
- 350 MB on disk
- 12+ s / image on CPU

Not well fitted for resource-constrained deployments.



Accuracy vs. size/latency.

[1] Enstad et al. (2025). *Comparative analysis of optical character recognition methods for Sámi texts from the National Library of Norway.* Proc. NoDaLiDa/Baltic-HLT, pp. 98–108.

Research question & contributions

Research question

Can a lightweight OCR architecture approach Transformer-level accuracy for North Sámi while enabling practical deployment on modest hardware?

What makes North Sámi hard for OCR

1. **Lack of publicly available real scanned documents**
2. **Seven special diacritics**
   á, č, đ, ŋ, š, ŧ, ž
3. **Agglutinative morphology**
   Hundreds of inflected forms per verb, large type-token ratio → model cannot memorise word shapes.
4. **Consonant gradation**
   giella ("language") vs. *giela* ("of language") – a single dropped "*l*" changes the meaning.

Northern Sámi Special Characters

Á/á *áhkku (grandmother)*	**Č/č** *čáhci (water)*	**Đ/đ** *ođđa (new)*	**Ŋ/ŋ** *eaŋgals (english)*
Š/š *šaldi (bridge)*	**Ŧ/ŧ** *máŧolaš (possible)*	**Ž/ž** *iežá (other)*	

Two architectural approaches for OCR

CRNN[2] **based OCR**: CNN (Convolutional Neural Network) + RNN (Recurrent Neural Network) + CTC (Connectionist Temporal Classification)

- Reads image from left to right
- CNN: extracts column features
- RNN: sequence encoder
- CTC: head + Loss, align length-varying input to output

Transformer based OCR (TrOCR)[3]: ViT (Vision Transformer) encoder + autoregressive decoder

- Image is split into patches and processed with self-attention
- Token-by-token generation based on the encoded image
- Pre-trained on massive image corpora

[2] Shi, Bai, Yao (2017). *An end-to-end trainable neural network for image-based sequence recognition and its application to scene text recognition.* IEEE Trans. PAMI, 39(11), pp. 2298–2304.

[3] Li et al. (2023). *TrOCR: Transformer-based optical character recognition with pre-trained models.* Proc. AAAI, 37, pp. 13094–13102.

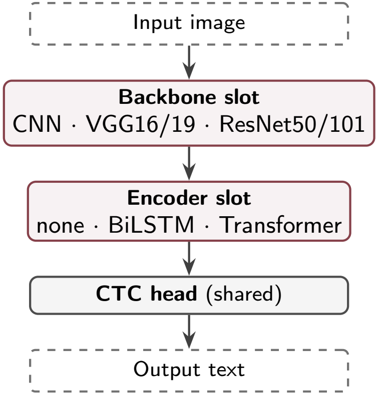
A modular OCR pipeline

Based on CRNN: 5 backbones × 3 encoders, shared CTC head.

Three swappable slots:

- Backbones: Feature extraction
- Encoders: Sequence Context
- Decoder: Shared CTC head

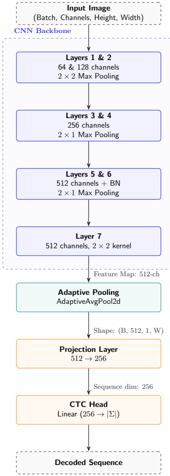
Due to compute budget only four variants were trained. Three failed to converge.



The headline model: CNN-CTC (ctc_simple)

7-layer CNN, trained from scratch

1. Preprocessing: Greyscale, normalization, resize
2. Feature extraction: 2x2 max pooling + 2x1 asymmetric pooling
3. No encoder: extracted features passed directly to CTC head (adaptive pooling + projection layer)
4. Sequence decoding: 395 classes, greedy decoding, collapses repeated characters and removes blank tokens.



Experimental setup

Train Språkbanken synthetic Sámi OCR dataset (National Library of Norway), ~307k line images (276k training + 31k hyperparameter tuning), 32×800px.

Evaluation 1,048 samples

- 1,000 in-domain synthetic lines (Språkbanken validation split)
- 48 out-of-domain parallel Northern Sámi–English sentences from Turi's 1910 "An Account of Sámi"

Metrics CER (Character Error Rate), WER (Word Error Rate), exact match, normalised accuracy, Character/Word Accuracy (100 – CER/WER)

Main benchmark results

Model	Char. Acc. % ↑	Word Acc. % ↑	sec / img	Size (MB)
trocr_smi_synth	**91.45**	**71.84**	12.41	~350
ctc_simple	87.76	62.65	**0.033**	**23**
trocr_smi_pred_synth	84.15	54.24	12.08	~350
trocr_smi	75.88	36.50	16.23	~350

$\textit{Char. Acc.} = 100 - CER \quad \textit{Word Acc.} = 100 - WER$

Results for `ctc_simple`**:** 87.76% char. acc. (within 3.69 pp of best TrOCR), 380× faster, 15× smaller.

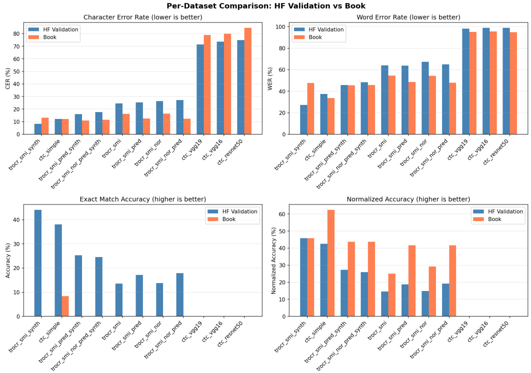
Throughput: 109,000 img/hour for `ctc_simple` vs 290 img/hour for TrOCR
Real-life: Scanning 1 milion of pages takes <10 hours on CPU vs. ~144 days for TrOCR.

In-domain vs Out-of-domain evaluation

On 1910s historical text (Friis ortography):

- TrOCR degrades 4.84 pp
- `ctc_simple` accuracy stays flat
- On normalised accuracy `ctc_simple` performs better

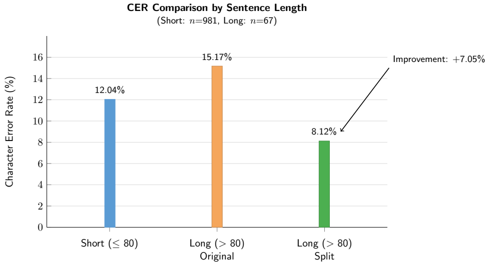
But!
$n = 48$ samples. Not conclusive, might suggest...
The CNN-CTC is more stable but results lack statistical power.



Conjunction-based sentence splitting

- Sentences with over 80 characters have higher CER.
- Mitigation: Split at North Sámi conjunctions (*ja*, *muhto*, *dahje*, *go*, *ahte*, *jos*), OCR each segment, rejoin.
- Long-sentence CER drop: -7.05 pp

Proof of concept: operates on ground-truth, the cut images are re-rendered without noise.



End-to-end OCR → MT pipeline

- CNN-CTC output is feed to TartuNLP API (sme to eng)
- Translation accuracy is moderate
- Character errors land on parts carrying gramatical role
- Useful for indexing content discovery but not full translation

Metric	Score
BLEU ↑	10.40
chrF ↑	33.85
TER ↓	76.30%

BLEU (Bilingual Evaluation Understudy) chrF (character n-gram F-score) TER (Translation Edit Rate)

Example (CER 1.2%):

GT (sámi): *ja áldoeallu lea váralaš dan galggalii sáhttit nu siivvut reainnidit go lea máŧolaš*

Pred (sámi): *. . . sáhttit nu siivut reainnidit . . . [one "v" dropped]*

GT (en): "and it is a dangerous time for the cow herd, they must be herded as carefully as possible."

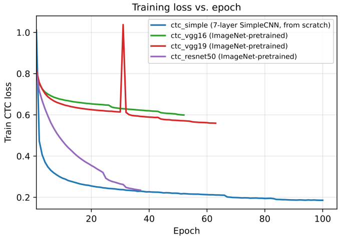
Pred (en): "and the livestock is dangerous, it should be able to be removed as safely as possible."

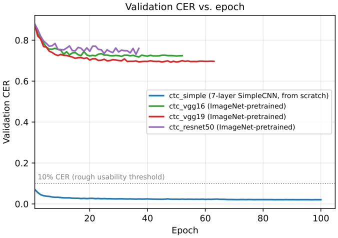
Pretrained backbones failed training

Three ImageNet pretrained backbones failed to converge.

- VGG16/VGG19: *underfitting* - the training loss barely moved
- ResNet50: *overfittng* - the training loss went down but validation loss didn't move

Hypothesis: ImageNet's weights were pre-trained to detect object-level features and not character-level.





Conclusion

Were the goals achieved?

- Lightweight CNN-CTC OCR model for North Sami, achieving 87.76% character accuracy, within 3.69 pp CER of state of the art, 380× faster and 15 × smaller with stable historical-text behaviour
- Makes the deployment easier for communities, which the technology should serve.

Applications & future work

Applications

- Mobile / edge OCR for community digitalization
- Large (milions) scanned documents indexing
- Joint OCR + reconstruction of occluded text (Figure)

Future work

- Real scanned corpus
- Giellatekno morphological post-correction
- Layout analysis for full pages
- Transfer to other Sámi varieties (Inari, Skolt, Lule)

Damaged-text reconstruction proof of concept: OCR + diffusion inpainting on an artificially occluded line.

Thank you

Questions?

`jamus23@student.sdu.dk`
`github.com/magwrap/north-sami-ocr`
`huggingface.co/magwrap/cnn-ctc-ocr-sme`